

A parity-separated partial result for anti-concentration under unconditional log-concavity

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Status. This is a `partial_result_likely_valid` packet for arXiv:2603.22664. It proves a structural subcase of Conjecture 1.4 in Glazer–Mikulincer, and does not claim the full conjecture.

Source question. In the discussion after Theorem 1.3, Glazer–Mikulincer ask for uniform lower bounds on the L^2 norm of degree- d homogeneous polynomials under isotropic log-concave measures invariant under signed coordinate permutations. In their Conjecture 1.4, with the implicit normalization $\text{coeff}_d(f) = 1$, they predict a constant $C_d > 0$ such that

$$\int_{\mathbb{R}^n} f^2 d\mu \geq C_d$$

for every isotropic log-concave H_n -invariant measure μ and every degree- d homogeneous polynomial f . Question 1.5 asks whether the sharp constant is governed, up to $O(1/n)$, by the isotropic cube.

Definitions. For a multi-index $I = (I_1, \dots, I_n)$, write $x^I = \prod_j x_j^{I_j}$ and $|I| = \sum_j I_j$. A set A of degree- d multi-indices is *parity-separated* if for every distinct $I, J \in A$ there is a coordinate k such that $I_k + J_k$ is odd, equivalently $I \not\equiv J \pmod{2}$ coordinatewise.

Proof intuition. The obstruction in the source conjecture is cancellation between monomials that transform according to the same sign character. If this cannot happen, unconditionality does all the representation theory for free: after conditioning on the absolute values $|X_i|$, the independent random signs kill every cross term. The problem reduces to proving that each fixed monomial has nontrivial second moment. That is a finite-dimensional consequence of log-concavity: each one-dimensional isotropic log-concave marginal has uniformly bounded density, so a fixed set of at most d coordinates is simultaneously bounded away from zero with probability bounded below in terms of d only.

Lemma 1 (Fixed monomial lower bound). *There is a constant $m_d > 0$ depending only on d such that the following holds. Let X be an isotropic log-concave random vector in $\mathbb{R}^n$. If I is a multi-index with $|I| = d$, then*

$$\mathbb{E}|X^I|^2 \geq m_d.$$

Proof. Let $S = \{j : I_j > 0\}$, so $1 \leq |S| \leq d$. A standard one-dimensional fact says that every isotropic log-concave real random variable has density bounded above by a universal constant L . Hence, for every coordinate j ,

$$\mathbb{P}(|X_j| < a) \leq 2La.$$

Choose $a = (4Ld)^{-1}$. By the union bound,

$$\mathbb{P}(|X_j| \geq a \text{ for all } j \in S) \geq 1 - 2L|S|a \geq \frac{1}{2}.$$

On this event, $|X^I|^2 = \prod_{j \in S} |X_j|^{2I_j} \geq a^{2d}$. Therefore $\mathbb{E}|X^I|^2 \geq a^{2d}/2$, and the right-hand side depends only on d . $\square$

Theorem 1 (Parity-separated unconditional lower bound). *For every $d \in \mathbb{N}$ there is a constant $c_d > 0$ such that the following holds. Let μ be an isotropic unconditional log-concave probability measure on $\mathbb{R}^n$. Let*

$$f(x) = \sum_{I \in A} a_I x^I$$

be a degree- d homogeneous polynomial whose support $A \subseteq \{I : |I| = d\}$ is parity-separated. Then

$$\int_{\mathbb{R}^n} f(x)^2 d\mu(x) \geq c_d \sum_{I \in A} a_I^2.$$

In particular, if $\text{coeff}_d(f) = 1$, then $\int f^2 d\mu \geq c_d$.

Proof. Let $X \sim \mu$. Since μ is unconditional, we may write $X = \varepsilon \odot R$, where $R = (|X_1|, \dots, |X_n|)$ and, conditionally on R , the signs $\varepsilon_1, \dots, \varepsilon_n$ are independent Rademacher variables. Then

$$\mathbb{E}[f(X)^2 | R] = \sum_{I, J \in A} a_I a_J R^{I+J} \mathbb{E}[\varepsilon^{I+J}].$$

The sign expectation is zero unless $I \equiv J \pmod{2}$. Since A is parity-separated, the only surviving terms are the diagonal terms. Thus

$$\mathbb{E}[f(X)^2 | R] = \sum_{I \in A} a_I^2 R^{2I}.$$

Taking expectations and applying Lemma 1 to each I gives

$$\mathbb{E}f(X)^2 = \sum_{I \in A} a_I^2 \mathbb{E}|X^I|^2 \geq m_d \sum_{I \in A} a_I^2.$$

Set $c_d = m_d$. $\square$

Corollary 1 (Multilinear subcase). *Conjecture 1.4 of Glazer–Mikulincer holds for all degree- d multilinear homogeneous polynomials under the weaker assumption that μ is isotropic, unconditional, and log-concave. If f is multilinear and homogeneous with $\text{coeff}_d(f) = 1$, then*

$$\int f^2 d\mu \geq c_d.$$

Moreover, since every nonconstant multilinear monomial has an odd exponent in some coordinate, $\mathbb{E}f = 0$ under unconditional μ . Therefore $\text{Var}_\mu(f) \geq c_d$, and the equivalence Proposition 1.1 in the source paper gives dimension-free small-ball and Fourier decay with exponent $1/d$.

Limitations. The theorem deliberately avoids the same-parity blocks. In those blocks, conditional sign averaging leaves cross terms $a_I a_J |X|^{I+J}$, so cancellations may persist. This is precisely where the source paper's higher-degree difficulty lives, especially for even symmetric polynomials such as functions of $x_1^2, \dots, x_n^2$. The result also does not address the cube-extremal constant in Question 1.5; the constant produced above is only a robust dimension-free lower bound.

Novelty check. The local run indexes had no entry for arXiv:2603.22664. A bounded web search on 2026-07-18 for phrases including “unconditional log-concave multilinear polynomial small ball variance lower bound”, “isotropic unconditional log-concave homogeneous polynomial L2 norm coefficient lower bound”, and “hyperoctahedral invariant log-concave homogeneous polynomial L2 norm conjecture” found the source paper and product-measure / L^p -ball results, but no exact parity-separated unconditional diagonalization statement.

Human-review recommendation. Check the standard one-dimensional density bound used in Lemma 1 and the conditional random-sign representation for unconditional log-concave measures. The rest of the proof is a direct orthogonality computation.

References

- [1] I. Glazer and D. Mikulincer, *Anti-concentration of polynomials: L^p balls and symmetric measures*, arXiv:2603.22664.
- [2] L. Lovasz and S. Vempala, *The geometry of logconcave functions and sampling algorithms*, Random Structures Algorithms 30 (2007), 307–358.